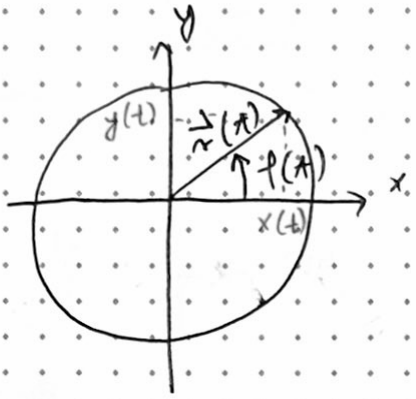
F 2



$|\vec{r}(t)| = r = \text{konst.}$

(Definicija krožnja)

$$x^2 + y^2 = r^2$$

$$x(t)^2 + y(t)^2 = r^2$$

$$\varphi(t) \rightarrow \begin{cases} x(t) = r\cos\varphi(t) \\ y(t) = r\sin\varphi(t) \end{cases} \Big\} \vec{r}(t)$$

$$z(t) = 0 \quad (\text{ker smo v } 2d)$$

odvod lege po času

① $\vec{v}(t) = \dot{\vec{r}}(t) = (\dot{x}, \dot{y}, \dot{z})$

$\vec{v}(t) = (r(-\sin\varphi(t))\dot{\varphi}, r(\cos\varphi(t))\dot{\varphi}, 0)$

② $\vec{a}(t) = \dot{\vec{v}}(t) = (\dot{v}_x, \dot{v}_y, \dot{v}_z)$

$= \underbrace{r\dot{\varphi}(t)}\underbrace{(-\sin\varphi(t), \cos\varphi(t), 0)}$

$(f\cdot g)' = f'\cdot g + f\cdot g'$

$\alpha(t)$

$\vec{a}(t) = r\boxed{\ddot{\varphi}(t)}(-\sin\varphi(t), \cos\varphi, 0) +$

$+ r\dot{\varphi}(t)((-\cos\varphi(t)\dot{\varphi}(t), (-\sin\varphi(t))\dot{\varphi}(t), 0)$

$$a(t) = -|\dots| + \left(-r\,\underbrace{(\dot{\varphi}(t))^2}_{=\omega(t)^2}\right)(\cos\varphi(t), \sin\varphi(t), 0)$$

KOTNA HITROST

$$\omega = \dot{\varphi}(t) \quad [\text{rad/s}]$$

KOTNI POSPEŠEK

$$\alpha = \dot{\omega} = \ddot{\varphi}$$

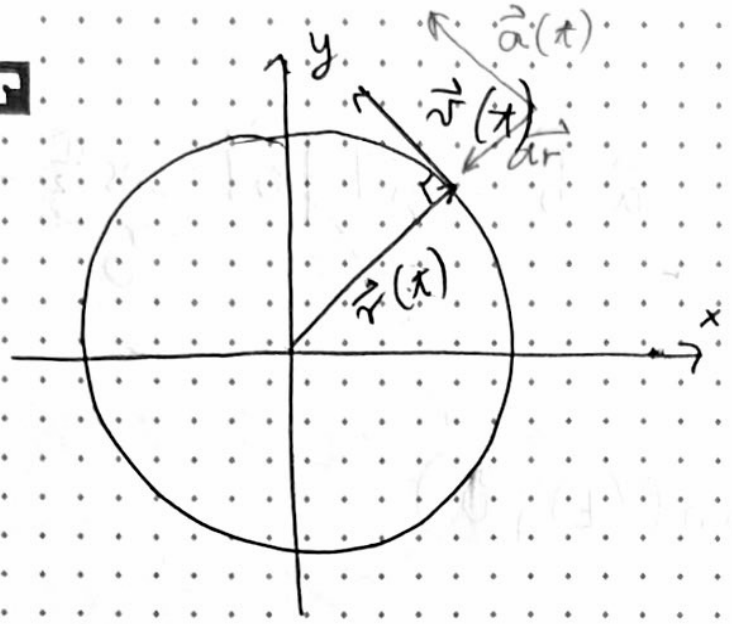
① ENAKOMERNO KROŽENJE

$$\left.\begin{aligned} \omega &= \text{konst} \\ \alpha &= 0 \end{aligned}\right\} \quad \varphi(t) = \varphi_0 + \omega t$$

② ENAKOMERNO POSPEŠENO K.

$$\alpha = \text{konst} \Rightarrow \omega = \omega_0 \cdot t\,\alpha \cdot t$$

$$\varphi(t) = \varphi_0 + \omega_0 t + \frac{\alpha t^2}{2}$$



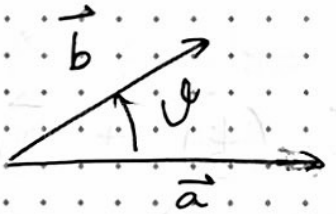
~~$\vec{v}(t) = r \cdot \omega$~~

$$\vec{v}(t) = r \cdot \omega \left(-\sin\varphi(t), \cos\varphi(t), 0 \right) \dots$$

↳ SKALARNI PRODUKT

$$s = \vec{a} \cdot \vec{b}$$

$$s = |\vec{a}| \cdot |\vec{b}| \cos\vartheta$$



$$s = a_x \cdot b_x + a_y \cdot b_y + a_z \cdot b_z$$

$$s = \sum_{i=1}^{3} a_i \cdot b_i \qquad (a_1 = a_x, \dots)$$

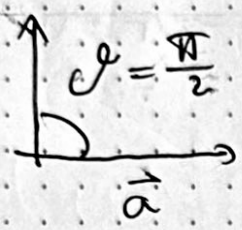
(neta) $\vartheta = 0$

$$\Downarrow$$

$$|\vec{a}|^2 = \vec{a} \cdot \vec{a} = |\vec{a}| \cdot |\vec{a}| \cdot \cos\vartheta \qquad \uparrow = 1$$

$$\Rightarrow \cancel{v(t)} \quad v^2(t) = \vec{v} \cdot \vec{v} = (r\omega)^2 \underbrace{\left((-\sin\varphi)^2 + (\cos\varphi)^2 + 0 \right)}_{=1}$$

$$|\vec{v}(t)| = v(t) = r \cdot \omega(t)$$



$$s = \vec{a}\,\vec{b} = |\vec{a}|\,|\vec{b}| \underbrace{\cos\tfrac{\pi}{2}}_{0} = 0$$

$$\vec{r} = r(\cos\varphi(t), \sin\varphi(t), 0)$$

$$\vec{r}\cdot\vec{v} = r^2 \cdot \omega(\cos\varphi(-\sin\varphi) + \sin\varphi\cos\varphi + 0)$$
$$= 0 \quad (\text{kot prej sta pravokotna})$$

$$\vec{a}(t) = \vec{a}_t(t) + \vec{a}_r(t)$$
$$\vec{a}_t \parallel \vec{v}$$

TANGENTNI POSPEŠEK

radialen pospešek ↓

$$a_t(t) = |\vec{a}_t| = r \cdot \alpha(t) \leftarrow \vec{a}_r(t) = -\omega^2\vec{r}$$
$$, \vec{a}_r \parallel (-\vec{r})$$

$$a_r(t) = |\vec{a}_r(t)| = \omega^2 r$$

ZGLED: kako je z pospeški pri enakomernem kroženju

$\vec{a}$ za $\omega = \text{konst}$?

$\dot{\omega} = \alpha = 0 \Rightarrow \vec{a}_t = 0$

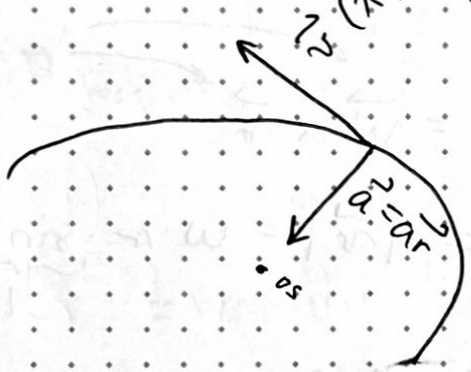
↳ $a_r = \omega^2 r = \text{konst}$

↳ $\vec{a}_r = -\omega^2\vec{r}$ (spreminja smer)

↳ $v = |\vec{v}(t)| = \omega r = \text{konst}$

$\vec{v}$ → spreminja smer

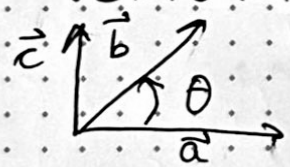
hecna lekcija - enakomerno kroženje je še vseeno pospešeno gibanje (ar kaže proti sredini)



čim krožiš se smer hitrosti spreminja → tvoj je pospešeno gibanje (zato smo na zemlji v pospešenem sistemu)

POENOSTAVLJENO:

VEKTORSKI PRODUKT

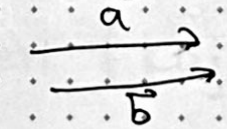


$$\vec{c} = \vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$$

$$|\vec{c}| = |\vec{a}||\vec{b}| \sin\theta$$

$$\Rightarrow \theta = 0 \Rightarrow |\vec{c}| = 0$$

smer definirana s pozitivnim pravilom



ni vektorskega produkta (ni ravnine)

$$\Rightarrow \theta = \frac{\pi}{2} \Rightarrow |\vec{c}|_{max} = |\vec{a}||\vec{b}|$$

ZVIT TRIK ↗ gor če poz smeri
dol če neg smeri

$\vec{v} = \vec{\omega} \times \vec{r}$ ← $\theta = \frac{\pi}{2}$

$v = |\vec{v}| = \omega r \underbrace{\sin\theta}_{1}$

$v = \omega r$

$\vec{\omega} \perp \vec{v}$

$\vec{v} \perp \vec{r}$

$\vec{a} = \dot{\vec{v}} = \underbrace{\dot{\vec{\omega}} \times \vec{r}}_{\vec{a}_t} + \underbrace{\vec{\omega} \times \dot{\vec{r}}}_{\vec{a}_r}$

$= \underbrace{\alpha \times \vec{r}}_{\vec{a}_t} + \underbrace{\vec{\omega} \times \vec{v}}_{\vec{a}_r}$

NEWTONOVI ZAKONI 1687

zunanji vpliv: sprememba gibanja / deformacija

① Če je vsota sil na telo enaka nič, teb miruje ali se giblje premo enakomerno

inercialni sistem

② Velja: $\vec{F} = m \cdot \vec{a}$ (pogoj, da velja)

↳ masa: inercija (nasprotovanje spremembam)

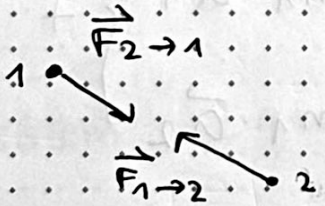
$\vec{F} = m \dot{\vec{v}}$

$\vec{F} = m \ddot{\vec{r}} \Rightarrow$ iščeš lego v odvisnosti od časa $\vec{r}(t)$

$\vec{F} = 0 \Rightarrow \vec{a} = 0 \Rightarrow \vec{v} = \text{konst}$

(premo enako gibanje)

③ Akcija – reakcija



$\vec{F}_{2\to1} = -\vec{F}_{1\to2}$

TRIKI

Newton: vpliv je takojšen (kaj pa med Zemljo in Soncem?)

obstaja odzivni čas sistema

TIPI TELES:

① Točkasto telo

⋮

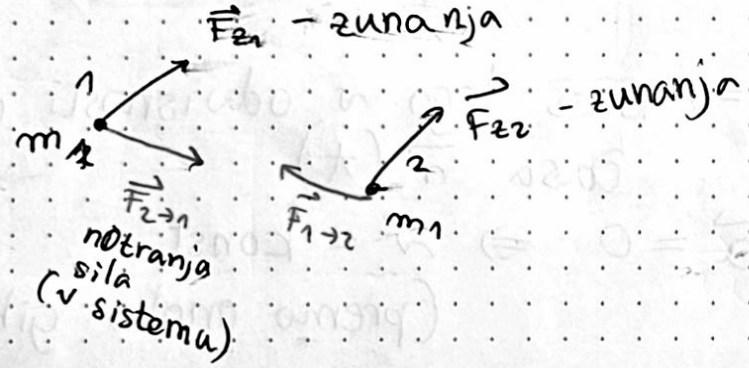
SISTEM VEČ TELES ~ TOČKAST

Osnovni gradnik snovi: atom $m_1 \neq 0$

↓

telo iz N atomov ⇒ sistem N teles

→ N.Z.: za N=2 atoma:



② N.Z.

1. telo $\vec{F}_1 = \vec{F}_{z1} + \vec{F}_{2\to1} = m_1 \cdot \vec{a}_1$ } +

2. telo $\vec{F}_2 = \vec{F}_{z2} + \vec{F}_{1\to2} = m_1 \cdot \vec{a}_2$ }

③ N.2.

$$\vec{F}_{2\to1} = -\vec{F}_{1\to2}$$

DOBIMO: $\vec{F}_{z1} + \vec{F}_{2\to1} + \vec{F}_{z2} + \vec{F}_{1\to2}$ (zunanji sili se odštejeta)

$= m_1 \cdot \vec{a}_1 + m_1 \cdot \vec{a}_2$

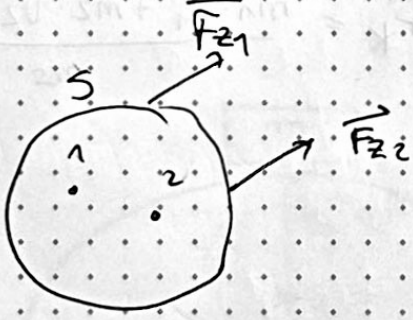
$\vec{F}_{z1} + \vec{F}_{z2} = m_1 \cdot \vec{a}_1 + m_1 \cdot \vec{a}_2$

$\vec{F}_z = \underbrace{2m_1}_{\text{masa sistema}} \cdot \underbrace{\frac{(\vec{a}_1 + \vec{a}_2)}{2}}_{\text{pospešek sistema}}$

$\vec{F}_z = m_{sistem} \cdot a_{sistema}$

NPR: $\vec{a}_1 = \vec{a}_2$

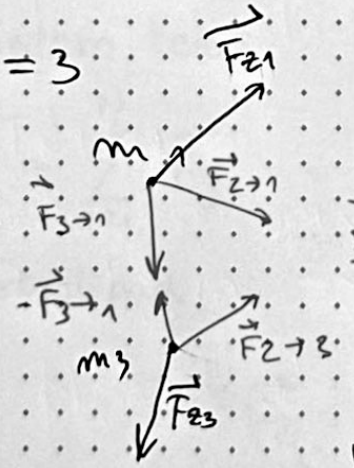
1 ——— 2 (R = fiksna) $\Rightarrow \vec{a}_2 = \vec{a}_1$

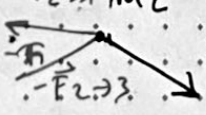


$\vec{a}_{sistema} \Leftrightarrow \vec{F}_z = \vec{F}_{z1} + \vec{F}_{z2}$

$m_{sistem} = 2m_1$

$N = 3$





junchline: vsota sil je vedno 0 (zaradi ③ N.Z.)

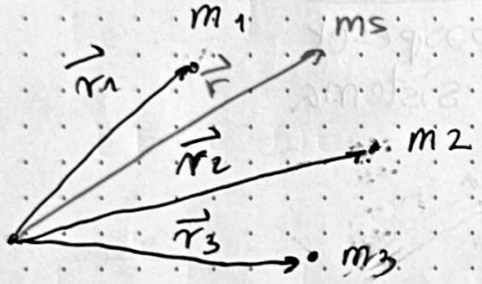
$\underbrace{\vec{F}_{z1} + \vec{F}_{z2} + \vec{F}_{z3}}_{\vec{F}_z} = m_1 \cdot \vec{a}_1 + m_2 \cdot \vec{a}_2 + m_3 \cdot \vec{a}_3$

$m_s = m_1 + m_2 + m_3$

$$\vec{a}_s = \frac{m_1 \cdot \vec{a}_1 + m_2 \cdot \vec{a}_2 + m_3 \cdot \vec{a}_3}{m_s \quad (m_1 + m_2 + m_3)}$$

↳ vemo: $\vec{a}_1 = \vec{a}_2 = \vec{a}_3$ (togo telo)

$$\vec{a}_3 = \vec{a}_1$$



$$\vec{r}_* = \frac{m_1 \cdot \vec{r}_1 + m_2 \cdot \vec{r}_2 + m_3 \cdot \vec{r}_3}{m_s}$$

$$\Downarrow$$

$$\vec{a}_s = \ddot{\vec{r}}_* = \frac{m_1 \cdot \vec{a}_1 + m_2 \cdot \vec{a}_2 + m_3 \cdot \vec{a}_3}{m_s}$$

$$\Downarrow$$

$$\vec{v}_s = \dot{\vec{r}}_* = \frac{m_1 \cdot \vec{v}_1 + m_2 \cdot \vec{v}_2 + m_3 \cdot \vec{v}_3}{m_s}$$

$\vec{F}_z = m_s \cdot \vec{a}_s$

($\vec{a}$ masnega središča - težišče)

center of mass

SISTEM N TELES

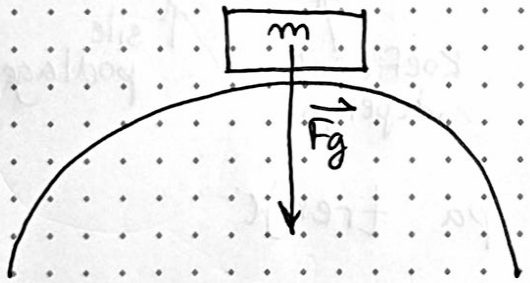
$$\vec{r}_* = \frac{\sum_{i=1}^{N} m_i \vec{r}_i}{m_s} \Rightarrow \vec{v}_s = \frac{\sum_{i=1}^{N} m_i \vec{v}_i}{m_s}$$

$$m_s = \sum_{i=1}^{N} m_i \Rightarrow \vec{a}_s = \frac{\sum_{i=1}^{N} m_i \cdot \vec{a}_i}{m_s}$$

$$\vec{F}_z = \sum_{i=1}^{N} \vec{F}_{zi}$$

$$\vec{F}_z = m_s \cdot \vec{a}_s$$

SILA TEŽE (sila gravitacije na Zemlji)



$$\vec{F}_g = m \cdot \vec{g} \quad [N = kg\,m/s^2 \text{ newton}]$$

$$\vec{g} \approx 9{,}8\ m/s^2$$

sistem teles:

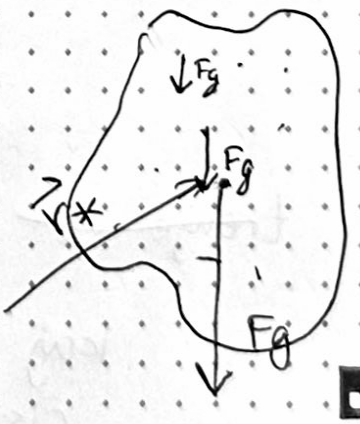
$$F_z = \sum_{i=1}^{N} \vec{F}_{gi} = \sum_{i=1}^{N} m_i \vec{g} = m_s \cdot \vec{g}$$

↑ prosti pad

$$\vec{F}_z = m_s \cdot \vec{a}_s$$

$$m_s \cdot \vec{g} = m_s \cdot \vec{a}_s$$

damo v masno središče



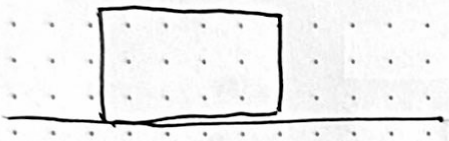
inercialna masa

$\vec{F_g} = m_g \cdot \vec{g}$ $\qquad$ $\vec{F} = m \cdot \vec{a}$

gravitacijska masa

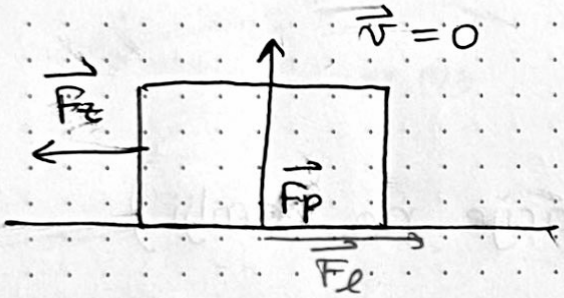
$mg = ma$?

TRENJE



→ nazobčenost površine

LEPENJE → TRENJE



$\vec{F_l}$ = sila lepenja

smer $(-\vec{F_z})$

$F_l^{max} = k_l \cdot F_p$

k_l ↑ koeficient lepenja

F_p ↑ sile podlage

ko se enkrat premika pa trenje

$\vec{v} \neq 0$

$F_{tr} = k_{tr} \cdot F_p$

$k_{tr} < k_l$

~~trenje je~~ jeklo na jeklo

kaj bi se zgodil če bi površine čisto zgladili? (proces varjenja)